

Bit Manipulation Ultimate Formula Sheet (Easy → Medium)

This comprehensive guide serves as a quick reference for technical interviews and competitive programming, covering fundamental bitwise operations, common patterns, and advanced tricks.

Core Bitwise Operations

Topic	Formula / Trick	Purpose
Check Odd	$(n \& 1) == 1$	Number is odd
Check Even	$(n \& 1) == 0$	Number is even
kth Bit Mask	$1 \ll k$	Create mask with only kth bit set
Check kth Bit	$(n \& (1 \ll k)) \neq 0$	Is kth bit set?
Set kth Bit	$n \mid (1 \ll k)$	Set kth bit
Clear kth Bit	$n \& \sim(1 \ll k)$	Make kth bit = 0
Toggle kth Bit	$n \oplus (1 \ll k)$	Flip kth bit
Multiply by 2^k	$n \ll k$	Fast multiplication
Divide by 2^k	$n \gg k$	Fast division

Power of 2 Patterns

Topic	Formula	Purpose
Power of 2 Check	<code>n > 0 && (n & (n - 1)) == 0</code>	Check if <code>n</code> is a power of 2
Remove Rightmost Set Bit	<code>n & (n - 1)</code>	Deletes lowest set bit
Extract Rightmost Set Bit	<code>n & (-n)</code>	Gets lowest set bit
Negative Number	<code>-n = (~n) + 1</code>	Two's complement representation
NOT Operator	<code>~n = -(n + 1)</code>	Quick calculation of bitwise NOT

Set Bit Counting

Topic	Formula	Purpose
Built-in Count	<code>Integer.bitCount(n)</code>	Java built-in function
Brian Kernighan's	<code>while(n!=0) { n &= (n-1); cnt++; }</code>	Efficient set bit counting
LeetCode 191	Number of Set Bits	Count 1s in binary
LeetCode 338	Counting Bits	DP + bit tricks for range [0, n]

XOR Mastery

Property	Formula	Description
Same Cancels	$a \oplus a = 0$	XORing a number with itself results in 0
Identity	$a \oplus 0 = a$	XORing with 0 returns the original value
Rearrangement	$a \oplus b \oplus a = b$	Commutative and associative
Commutative	$a \oplus b = b \oplus a$	Order does not matter
Associative	$(a \oplus b) \oplus c = a \oplus (b \oplus c)$	Grouping does not matter

XOR Problem Recognition

Problem Type	Trick
One unique, others twice	XOR all elements
Missing Number	XOR indices + values
Hamming Distance	<code>bitCount(x ^ y)</code>
Two Unique Numbers	<code>xor & (-xor)</code> to partition into two groups
One Unique, others thrice	Bit count modulo 3

Bitmasking & Subsets

Operation / Topic	Formula
Add element	<code>`mask</code>
Remove element	<code>mask &= ~(1 << k)</code>
Check element	<code>(mask & (1 << k)) != 0</code>
Toggle element	<code>mask ^= (1 << k)</code>
Generate All Subsets	<code>for(mask=0; mask < (1 << n); mask++)</code>
Check inclusion	<code>(mask & (1 << i)) != 0</code>
Number of Subsets	<code>2^n</code>
Subset Size	<code>Integer.bitCount(mask)</code>

Interview Pattern Recognition

If You See...	Think...
Odd / Even	<code>n & 1</code> ▾
Power of 2	<code>n & (n - 1)</code> ▾
Count bits	Brian Kernighan's Algorithm ▾
Unique element	XOR ▾
Missing number	XOR ▾

If You See...	Think...
Hamming distance	XOR + bitCount ▾
Two unique numbers	XOR + $n \& (-n)$ ▾
Generate all subsets	Bitmasking ▾
$n \leq 20$	Bitmasking candidate ▾
Pick / Don't Pick	Subsets / Bitmask ▾

Top 10 Must-Memorize Formulas

#	Formula
1	$n \& 1$
2	$n \& (1 \ll k)$
3	$\sim n$
4	$n \& \sim(1 \ll k)$
5	$n \wedge (1 \ll k)$
6	$n \& (n - 1)$
7	$n \& (-n)$
8	$a \wedge a = 0$
9	$a \wedge 0 = a$

#	Formula
10	<pre>for(mask=0; mask < (1 << n); mask++)</pre>